Causal Software Clock Discipline over Wireless Physical Events: Native Lock Evidence, Operating Tiers, and a Tunable-Oscillator Extension

Jonathan Yantis (Engineer)

Research preprint for local review — 3 October 2026

Abstract

This study separates software phase-lock status, causal clock-map error and independently qualified oscillator/output accuracy. An overnight nRF52840 experiment failed to sustain native estimator readiness; its final closure retains 71 captures, including 38 incomplete captures. A subsequent coherence fix and bounded affine-rate/phase-slew servo produced native software lock under an explicit raw-radio profile. Four UI5 cohorts remain separate. Quiet-pair captures gave uncentered earlier-delivered-map RMS 69.136 and 68.106 ns on 1144/1144 and 727/728 retained originals; delayed-join captures gave 66.510 and 72.717 ns on 1106/1106 and 383/383. The final extension retained all 3579 scoreable originals: its first approximately ten minutes gave RMS 231.105 ns, absolute P99 1209 ns and maximum 1413 ns, while its subsequent approximately 23 minutes 52 seconds gave 79.896, 219 and 313 ns. Native lock appeared on 64/128 and 256/281 sampled statuses, without implying uninterrupted lock. The final displayed 84 ns was a floored recent-history statistic, not a run-wide bound. These results motivate an optional, capability-scoped `SOFTWARE_PLL` operating tier. Primary-source review and dimensional analysis define a proposed sampled PI extension for calibrated VCXO/OCXO/DCO or synthesizer actuation, including delay, saturation, phase alignment and holdover. Current boards provide a numerical clock-map actuator only. Fixed path bias, electrical output, quartz/carrier discipline, total uncertainty, energy and field restart behavior remain unqualified.

1 Question and contribution

A wireless clock can fit event timestamps closely while failing to deliver a stable coordinate at the time an application asks for it. It can also display a lock indicator while the physical oscillator remains free running. Precision Time over BLE (PToB) therefore treats event identity, causal publication, native control state and output metrology as separate obligations. The present question is whether the overnight raw-counter experiment establishes a useful software phase/frequency operating mode, and what a tunable-oscillator implementation would additionally require.

The deployed actuator is a software map from a quartz-derived counter to the configured reference's raw-counter coordinate. It adjusts numerical rate and slews numerical phase. The operator's intended meaning of `LOCKED` is algorithmic virtual lock: the phase/frequency solution is sufficiently accurate and stable, under the declared control profile, to support fine crystal correction and phase locking when a suitable actuator is available. On these fixed-quartz boards the solution controls the software map. No demonstrated interface pulls the quartz, and the experiment contains no RF carrier-phase detector. Actuator capability, actual engagement and independently qualified output accuracy remain separately reported properties. Readiness for a particular physical actuator additionally requires its calibration and closed-loop validation; algorithmic lock does not silently supply those results.

The contributions are: a retained-failure explanation tied to the final overnight closure; reproducible per-capture causal-error and sparse-state populations for four later cohorts; a precise account of the implemented affine feed-forward/slew method and its two independent dwell mechanisms; and an explicitly proposed hardware-control design with dimensional gains and validation gates. The failed simultaneous three-unit cohort

and final recovery tails are part of the result. Descriptive improvement across differently configured or situated cohorts does not isolate a causal RF treatment effect.

This manuscript extends the project's previous calibration/publication study [1]. That study prospectively froze a 10,516 ns ADDRESS-edge profile before PPS-shadow validation. The present raw-counter experiment has a different reference coordinate, estimator and score. Its learned intercept absorbs an unknown combined path/capture offset. The earlier calibration and PPS-shadow results cannot silently qualify this experiment's total phase error.

2 Primary-source methods and comparison limits

2.1 Software discipline and discrete control

The NTP reference specification describes a software variable-frequency oscillator with PLL/FLL discipline and a state machine that responds to offset, frequency, jitter and wander [2]. This is a precedent for clock-map control, not a source of interchangeable loop gains. Its poll-time scaling and error units must remain attached to its implementation.

LinuxPTP's primary PI implementation estimates initial frequency from separated offset samples, limits frequency commands and conditionally updates its integrator when the command is unsaturated. Its interval-dependent gains and stable-offset state illustrate how controller and lock semantics can be explicit [3]. Our installed method instead uses affine frequency estimation and bounded phase slew; the PI design below is a proposed alternative.

Tosato et al. derive a discrete PI servo for a programmable timer, with an explicit clock-generator transfer function and gain-stability region. Their wired-PPS/PRU implementation and normalized timer units differ from irregular radio events and delayed correction availability [4]. Freris, Borkar and Kumar develop stochastic clock models and asynchronous estimation under specified noise assumptions [5]. Neither paper's gains or measured performance are transferred to the present devices. We derive our proposed ideal sampled-loop conditions explicitly.

2.2 Wireless clock recovery versus packet clock maps

The Park–Wolverton patent describes reciprocal physical timestamps and a wireless filter controlling a tunable crystal, with separate phase/frequency considerations [6]. It supplies an architectural reference, not experimental evidence for these boards or an opinion on patent scope.

The AirClock poster extracts the difference frequency of two received tones through nonlinear processing and filtering, supplying a common hardware reference [7]. RFClock combines two-tone frequency recovery with a synthesizer, UWB time-of-flight, time-to-digital conversion and programmable delay for alignment. Its reported frequency offset below 0.107 Hz and phase deviation within 5 ns concern that hardware system; its separate 20 ps RMS PPS-period-jitter result is another statistic [8]. A packet-derived software map cannot inherit those figures or their physical carrier-coherence claim.

2.3 BLE precedents and measurement classes

CheepSync uses prior-transmission timing for passive BLE reception. BlueSync's author's dissertation describes radio-event PPI capture, subsequent-packet timestamps and physical output comparisons [9, 10]. Nordic's reference synchronization design also distinguishes radio capture from an observed output pin [11]. These precedents preclude treating follow-up timestamping or passive reception as uniquely new.

MicroSync evaluates connected nRF52840 devices through an external 800 MHz GPIO instrument and MCU power measurement. Its high-accuracy MAE is 272 ns, reported SD 319.2 ns and 99% CDF 591.25 ns; it does not report our causal-map RMS statistic [12]. Our smaller quiet-cohort numerical RMS is not a matched external-output or energy superiority result. The transport, detector, actuator, reference and population all differ.

An affine innovation scores a pre-existing estimator. A delivered-map prediction scores the pre-existing actuator coordinate. Remaining correction is a controller state. Own-PPS source prediction checks the local source model. Compare rounding checks a local scheduling mechanism. An independent output instrument checks the delivered electrical edge. Only the last, with a calibrated reference/path and full availability coverage, can support an electrical-output guarantee. No single one substitutes for the others.

3 Evidence closure and experimental population

All analysis is offline. The installed UI5 T096 revision is `28ee7d8`; the full identity and method hashes are in the evidence manifest [19]. Concurrent uninstalled IEEE 1588/BMCA edits are outside this experiment. Native V4 analysis uses the frozen deployed-method copy; native-only V3 analysis explicitly disables counterfactual replay. Logger USB arrival times are not a radio-event clock.

Each seal retains the run, capture configuration, installed-identity evidence, reviewed indexes, originals and complete/incomplete tails. A pre-seal analysis index is immutable: its statement that the run was not yet sealed describes its creation time. The later catalog and seal establish closure. The final seal digest is `99be920f8f09febce268874c72b9b6173e5cfa6fc711802c72e429fcc9b574bf`. These are private same-host archives, not independent durable backups or a public dataset.

The final run closed four files, retaining 2023 TC01 and 2010 TC03 own-PPS records plus 3579 TC01 pair records. Of the latter, 3543 match the original and all three TX-stage logs; 36 match the original with incomplete TX-stage logs. No additional original is inferred from metadata repetition. Two unterminated console tails remain parse-error inventory entries. The logger rotates first after ten minutes and subsequently schedules a one-hour rotation; closure produced ten minutes plus about 23 minutes 52 seconds, not repeated ten-minute segments.

TC01/TC03 silicon, runtime profiles and protected prewrite regions were verified. Fresh postwrite byte readbacks were deferred by explicit operator instruction. TC02's last bench status was native `PLL_LOCKED/CONVERGED`, 71 ns floored causal RMS and settled green zero; its field route, distance, departure instant and restart behavior were not measured, and no field USB record supplies a field-error population. Reset clears its volatile PLL and restores normal GM candidacy; automatic following on reset is not implemented. TC03 is the observed public GM and raw reference in the final cohort, while TC01 follows. Public GM identity and the configured diagnostic raw reference are distinct concepts.

4 Installed detector, estimator and actuator

4.1 Physical original and availability

The experiment uses raw-radio originals and low/high timestamp pages, separately from the supported-controller precision adapters. An original is identified by source, session, sequence, physical channel and captured event. Channel 37 originals supply the numerical observations. One reference transaction contains nine PDUs: ORIGINAL, LOW and HIGH on each of channels 37, 38 and 39. The next original is scheduled from the actual previous original with a 250 ms base and 0–10 ms offset; follow-up stages arrive roughly 20–30 ms later. Scheduling, repeated metadata and USB reporting are not additional phase observations.

Bounded pending and tombstone tables each have 16 entries, with a three-second completion deadline. Conflicts, retired sessions, invalid authority, malformed information and stale originals cannot refresh precision. The

low/high data must refer to the captured original; the carrier page's channel does not replace the original channel. Hardware RX time and correction-availability time are both recorded. This distinction prevents a later correction from improving an earlier delivered-clock score retroactively.

Let X_k be nominal nanoseconds for the reference's actual original TX tick, x_k the follower's physical RX tick and d_k the follower tick at which its correction becomes available. For the earlier delivered map M_{k-1},

$$e_k = X_k - M_{k-1}(x_k), \qquad d_{k-1} \le x_k. \tag{1}$$

Positive e_k means the delivered coordinate lags the reference event coordinate. A newly corrected map M_k cannot answer this historical test. The prior affine estimator's innovation is recorded separately. Invalid prior output remains an unavailable row, rather than a zero error.

The detector measures alignment of raw-counter coordinates at associated radio events. TX/RX capture latency and propagation are not independently removed. Fixed bias and total bound remain `UNKNOWN`. Receiver GNSS/PPS does not train the follower loop, and the reference's nominal raw tick conversion is not an absolute UTC discipline claim.

4.2 Affine frequency feed-forward with bounded phase slew

The installed estimator retains at most 32 fit points over at most 96 s, requires at least eight points and ten seconds of span, and thins by physical time at least 0.5 s apart. It does not select points by residual size. Its relative rate is clamped to ± 50000 ppb. A large original gap above five seconds resets the fit; a gap above three seconds prevents precision readiness.

At correction availability, the servo evaluates the old delivered coordinate at the current tick, reanchors there, applies the new fitted rate, and records the target-minus-delivered remaining phase. In ideal real arithmetic its map can be written

$$M(x) = A + q(x-D)(1 + r\,10^{-9}) + \operatorname{sgn}(P)\min\{|P|, Sq(x-D)10^{-9}\}, \tag{2}$$

where $q = 62.5$ ns/tick, D is the delivery anchor tick, A its delivered coordinate in ns, r the fitted rate in ppb, P the remaining phase in ns and $S = 1000$ ns/s. The expression is for $x \ge D$; the firmware uses bounded integer arithmetic. The 10^{-9} converts nominal elapsed ns to seconds in the slew term. At a valid update, reanchoring preserves continuity; rate and remaining phase change prospectively. Initial creation has no earlier delivered-output history. This is frequency feed-forward plus bounded phase slew, rather than a deployed PI integrator.

The nominal 16 MHz counter grid is 62.5 ns. Sub-tick means and regression coefficients are numerically meaningful statistical quantities, but they do not create a finer electrical edge grid. Delayed correction can still alter a future software query continuously; it cannot repair missed physical output events or retrospectively relabel their availability.

4.3 The overnight failure and coherence repair

The authoritative failed-V3 closure comprises 71 closed captures (33 complete, 38 incomplete), 261 original files and 558,440,887 raw bytes. It contains 7991 TC01 pair observations, 34,252 TC01 and 16,738 TC03 PPS rows, and 6279/8220 native status rows. No native `phase_ready/TRACK` is established. The final TC01 snapshot has fit count zero, 2739 resets and 174,469 invalidations; TC03 has 308,792 invalidations. These are cumulative snapshot counters, not per-capture loss probabilities.

Host probes reproducibly identify a coherence race capable of this failure: the caller's earlier "now" could precede a newly captured sample, causing a future-sample permission rejection and repeated invalidation. Native V3 did not log withdrawal reasons, so individual hardware invalidations cannot be attributed to that mechanism.

V4 snapshots the associated observation first, then samples current time, preserving coherent age/permission evaluation. It retains the radio timing, fault and channel guards. The initial failed V4 attempt and the subsequent finite/native cohorts remain separate sealed populations; later V4 captures do demonstrate native lock.

An earlier, smaller live closure of 50 captures supported an optimistic replay with permission enforcement omitted: 311 causal predictions among 5839 originals gave RMS 114.209 ns, but never reached a 32-bucket strict history. That is a counterfactual mechanism diagnostic on a different population. It is neither the final 71-capture native result nor a deployed precision success.

5 Native lock, quality histories and presentation

The experiment's permissive attempt gate requires at least eight fresh physical-time buckets over seven seconds, fit readiness, a four-update streak, original gap at most three seconds, and causal RMS strictly below 1000 ns. It allows controlled experimental tracking; it is not the strict lock criterion or an electrical acceptance threshold.

The strict estimator and delivered-output histories each require 32 fresh physical-time buckets spanning at least 30 s and the exact mean-square inequality

$$\sum_{i=1}^{n} e_i^2 < n(200\,\mathrm{ns})^2. \tag{3}$$

All innovations within a retained bucket contribute to its error sum, while the buckets prevent many clustered observations from impersonating long elapsed history. The displayed integer RMS is floored; threshold evaluation uses the exact sum of squares, not that floor.

Native `PLL_ACQUIRE`, `PLL_TRACKING` and `PLL_LOCKED` reflect actual permitted servo updates. Lock entry requires both strict histories, remaining phase magnitude at most 125 ns, delivered-error trend magnitude at most 10 ppb, and at least ten seconds plus four actual updates of qualifying dwell. Exit uses phase above 250 ns or trend above 20 ppb, with the strict-history, freshness, fault, authority and epoch obligations continuing. Missing or withdrawn permission, unavailable output, a changed epoch or a stale original clears qualification. A trend is fitted to 32 equally weighted signed bucket means at actual mean original times; its quantization rounds away from zero to avoid a falsely permissive truncated slope.

UI5's green zero is a separate five-second dwell requiring at least four advancing original sequences while native local lock and the correction prerequisites hold. The display's LOCK row and remaining-CORR value do not mean electrical zero error. Native convergence can precede settled green zero by about five seconds. The actual public GM always retains its PPB presentation; it does not turn into a follower solely because a diagnostic reference was configured.

Optional precision feedback reports peer state under a live authenticated control lease, but its payload itself is unsigned. Origin, configured peer, reference session, advancing original sequence and age below three seconds bind freshness. Duplicates cannot refresh or promote a state; they may downgrade it. Quantized RMS/error fields have unknown sentinels rather than a favorable clamp. UI5 adds an unsigned 32-bit remaining-slew magnitude with `UINT32_MAX` denoting unknown; signed phase/error telemetry is separate. Feedback does not train this loop, independently qualify the peer, or authorize public precision service.

The separation matters operationally. A reference in `LEADER/REFERENCE` may report its own inactive follower loop as `PLL_ACQUIRE/STALE_ORIGINAL` while remaining a valid raw reference. A fresh report that TC01 is locked is still peer-reported feedback, not TC03's own measured electrical follower lock.

6 Metric definitions and retained coverage

For the n scoreable retained originals in a named capture/epoch, we report

$$\bar{e} = \frac{1}{n}\sum_i e_i, \quad R = \sqrt{\frac{1}{n}\sum_i e_i^2}, \quad Q_p = \mathrm{sort}(|e|)_{\lceil pn \rceil}. \tag{4}$$

Bias is signed; RMS is uncentered. Absolute P95/P99 use nearest rank. No calibration refit, detrending, interpolation, phase recentering, outlier removal or favorable-lock-only subset is used. The unavailable denominator counts parsed retained originals that lack a causal delivered map. Logging drops mean this is not the denominator of all RF opportunities.

All tables use whole-capture native `hardware_window=-2` populations. Capture rotation is a reporting boundary, not a fit reset or a treatment. The original-gap distribution can include a prior observation across that boundary when the analyzer retained the same epoch. USB arrival gaps, missing sequence IDs and console counters cannot individually identify RF loss. State counts are observations at discrete status ticks; their ratios are not fractions of wall time or proof of continuous lock. Acquisition start can be left-censored at capture start; unobserved completion is right-censored at capture end. Status sampling interval-censors transitions.

The derivation first reads each review index, summary and epochs, then only indexed raw byte ranges for the pair/status plots. Every used raw file and reviewed analysis input is checked against its seal. It independently recomputes all seven UI5 error distributions and matches the native metrics within numerical tolerance. Companion CSVs preserve raw line, offset and length for each plotted point [19].

7 Results across separately retained cohorts

Table 1 names every UI5 TC01 capture and preserves its unavailable rows. The simultaneous three-unit cohort is substantially worse than the later pair and join cohorts. It retains 24 original gaps above three seconds, repeated fit resets and no qualifying native lock among 59 statuses. The later cohorts are descriptively better, but simultaneous activity, schedule history, geometry, joining order and estimator history differ; an isolated contention effect has not been established.

Table 1: Earlier delivered-map causal errors on all retained TC01 originals. Values are ns; n/N is scoreable/retained. No unlike captures are pooled.

Cohort / capture	n/N	Bias	RMS	P95	P99	Max. abs.
Simultaneous three / 1	354/384	−462.907	1075.679	2631	3072	3103
Quiet pair / 1	1144/1144	+9.698	69.136	134	174	222
Quiet pair / 2	727/728	+19.662	68.106	130	165	218
Delayed join / 1	1106/1106	+6.577	66.510	129	159	187
Delayed join / 2	383/383	+5.590	72.717	138	179	387
Final extension / 1	1062/1062	−15.271	231.105	382	1209	1413
Final extension / 2	2517/2517	+3.861	79.896	159	219	313

Quiet-pair capture 2 retains one unavailable delivered-map prediction among 728 originals. The delayed-join run's final snapshot is `PLL_TRACKING/ESTIMATOR_NOT_STRICT200`, despite many earlier locked snapshots and low whole-capture RMS. This demonstrates that a good aggregate distribution does not establish final readiness. TC02's bench lock is not included as a measured field cohort.

Table 2: Native qualified-lock and green-zero observations; gaps count retained-original intervals above three seconds. Status observations are a separate denominator from pair rows.

Cohort / capture	All statuses	Qualified lock	Green zero	Gaps > 3 s
Simultaneous three / 1	59	0	0	24
Quiet pair / 1	116	109	102	0
Quiet pair / 2	74	50	41	2
Delayed join / 1	118	93	85	2
Delayed join / 2	40	30	27	2
Final extension / 1	128	64	48	5
Final extension / 2	281	256	228	2

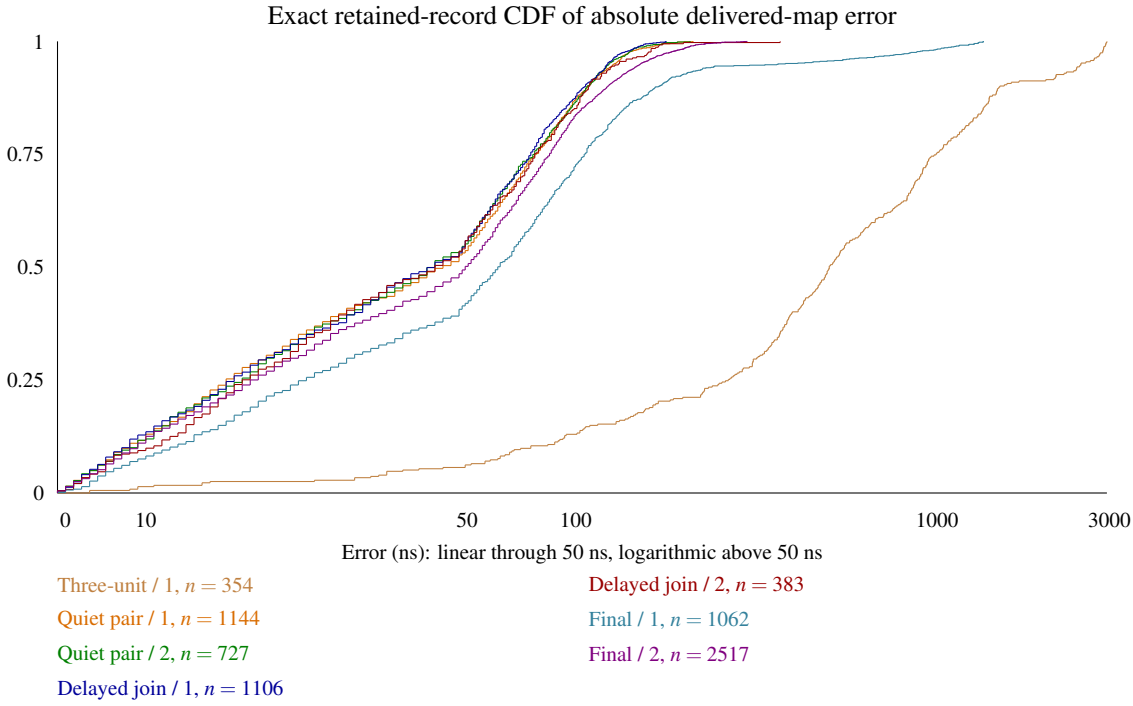


Figure 1: Exact empirical CDFs of the seven separate scoreable populations in Table 1. The linear-to-log transform preserves every tail; unavailable observations remain in the table denominator. The companion scientific exports and CSVs are generated by the same derivation.

8 Final extension: tails, recovery and delay

The first final capture starts at 17:07:24.937344 UTC and closes at 17:17:26.064518; the second starts at 17:17:26.065224 and closes at 17:41:19.122652 UTC. These are logger capture boundaries. Hardware event time, after subtracting an integer anchor, supplies the plot's elapsed axis. All errors, including the first capture's negative recovery excursion to -1413 ns, remain in Figure 2. A smaller final recent-window display cannot eclipse this interval.

At the last TC01 status, `PLL_LOCKED/CONVERGED` accompanies phase remaining zero, settled UI zero, trend 3 ppb, $n = 84$ and sum of squared delivered errors 597,462 ns^2. The exact recent RMS is $\sqrt{597462/84} = 84.336$ ns (rounded here); the device displays its floor, 84 ns. That population is neither 1062 nor 2517 nor 3579. Its zero is a convergence/UI state, not a calibrated zero physical-phase claim.

The first observed final native lock is 125.031 s after the first retained status; the first observed settled green zero is 130.070 s after that status. The loop was already operating when capture began, so these are time-to-first-observed states within this capture, not a power-cold acquisition benchmark. The first lock record is TC01 capture 1, line 1043, offset 534227, length 2494 bytes; the first green record is line 1087, offset 556011, length 2496. Twenty-one changes between adjacent observed state labels bracket transitions; status gaps prevent an exact transition instant or continuous lock-duration calculation.

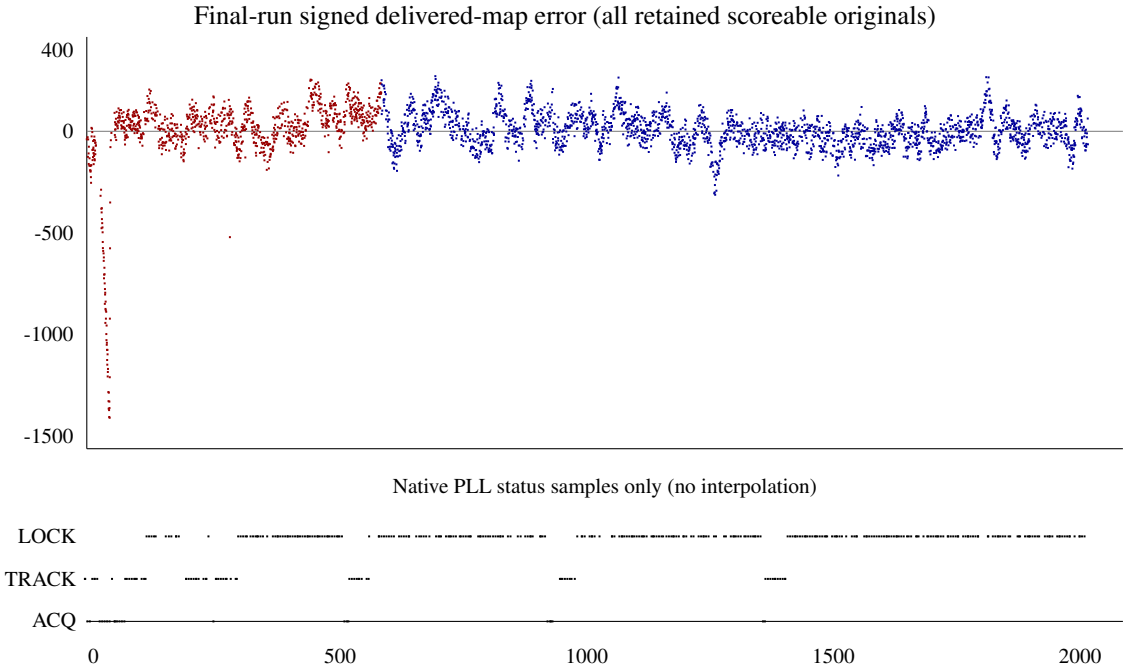


Figure 2: All 3579 final scoreable delivered-map errors against hardware elapsed seconds, with discrete native PLL observations below. Red is capture 1 and blue capture 2. The vertical range retains all observed errors. Status points are not connected and do not certify the intervals between them.

Table 3: Final retained-original gaps and correction availability. Latency is physical RX to completed correction processing, not USB transport.

Quantity	Capture 1	Capture 2
Gap intervals / retained originals	1061/1062	2517/2517
Mean / maximum gap (s)	0.5624 / 9.1899	0.5691 / 3.5498
Gaps above 3 s / above 5 s	5 / 2	2 / 0
Unobserved sequence IDs	726	1698
Correction latency min. / mean (ms)	66.144 / 191.459	80.698 / 206.848
Correction latency P99 / maximum (ms)	452.973 / 641.715	472.979 / 742.464

The second gap denominator includes the cross-rotation prior original. Gap and unobserved-sequence counts combine transport, reception, association, filtering and logging mechanisms; no RF-loss percentage is identifiable. The proposed controller must nevertheless tolerate the actual retained intervals and completion delay.

8.1 Diagnostic counters and distinct prediction classes

All sampled withdrawal-reason counter increments are zero. That does not erase retained gaps or state regression. Expiry and rejection include metadata/control processing and cannot be relabeled lost originals. The last

Table 4: Increments between the first and last sampled TC01 statuses in each final capture. These sampled counter intervals differ from the complete pair-record intervals.

Counter increment	Capture 1	Capture 2
Observations / affine predictions / unavailable	1052 / 1002 / 50	2521 / 2521 / 0
Duplicates / expired / rejected	59 / 735 / 493	143 / 1677 / 1217
Conflicts / malformed / invalidations	0 / 0 / 0	0 / 0 / 0
Fit resets	2	0
General log drops / PPS log drops	46 / 5	114 / 6
Missed internal compares	7	3

general/PPS drop counters include history before this run; only the named sampled increments are used here. No observer-free matched experiment measures whether logging altered reception, correction latency or oscillator behavior.

The first final capture's prior-affine innovation RMS is 85.182 ns on 1012 rows, with 50 unavailable affine predictions; its delivered-map RMS is 231.105 ns on all 1062 rows. The actuator retains a causal delivered coordinate during intervals in which the current fit is reacquiring, so these denominators and statistics differ legitimately. The second affine RMS is 79.441 ns on 2517 rows, alongside delivered RMS 79.896 ns. A low innovation alone cannot quantify what was delivered during recovery.

Own-PPS source predictions are separate local diagnostics. They use the local PPS/source-model path, not an independent cross-device electrical output. Frozen compare rows include the local timer's scheduling/rounding and a one-tick capture-minus-plan term. Neither trains the PLL or measures its GPIO pad, which is unconnected. This study makes no quartz, carrier, PPS-output or UTC accuracy inference from those records.

8.2 Cadence and resource interpretation

The current reference ceiling is four originals per second, each carrying nine PDUs. One reference and one follower can carry at most 42 precision-experiment PDUs/s; multiplying by the configured 376 μs PDU duration gives 15.792 ms/s, or 1.5792%. Adding TC02 feedback adds at most six PDUs/s and 2.256 ms/s, yielding 48 PDUs/s and 18.048 ms/s. These are configured carried-duration ceilings, not measured occupied RF airtime, receive listening duty, collision cost, CPU load or energy. Existing public/control traffic adds its own cost.

The cohorts suggest testing a fast acquisition cadence followed by slower maintenance, rather than assuming four originals per second are always needed. A credible comparison freezes firmware, authority, geometry, observation/logging profile and thermal history; alternates predefined rates; retains unavailable intervals and failures; and measures acquisition, maintenance error, completion probability, actual RX residence, CPU busy time and whole-device energy. Adaptive cadence must use genuine elapsed history and uncertainty growth, not favorable residual selection. Its savings and stability remain proposed.

9 Identifiability of phase, path and epoch

The radio detector cannot identify an absolute phase offset from an unknown one-way delay. In a common nominal time unit, let θ be follower-minus-reference clock phase, d the propagation delay and b the net TX/RX capture-path offset. A one-way measurement has the form

$$z = \theta + d + b + \eta. \tag{5}$$

The transformation $(\theta, d+b) \mapsto (\theta + c, d + b - c)$ leaves z unchanged. Repeated low-noise measurements or a small regression residual do not resolve this ambiguity. A time-varying path also contaminates a frequency

estimate: differentiating the measurement adds $\dot{d}+\dot{b}$ to apparent skew. A mobile link therefore needs a path model/measurement or an appropriately bounded uncertainty; timestamp trend is not automatically oscillator frequency error.

Reciprocal exchanges record four physical timestamps with explicit source/session/copy identity and turnaround timing. After skew and known capture-path corrections, define

$$z_f = \theta + D_{AB}, \qquad z_r = -\theta + D_{BA}, \quad \widehat{\theta} = \frac{z_f - z_r}{2} = \theta + \frac{D_{AB} - D_{BA}}{2}. \tag{6}$$

The familiar symmetric estimate removes a common delay but retains half the asymmetry. Different oscillator scales during turnaround must be normalized; otherwise clock-rate error enters the result. The patent's reciprocal architecture illustrates the need for such separate timestamp/frequency accounting [6]. Our equation is a simplified timing-identifiability analysis, not a reimplementation or validation of that patent.

If corrected residual delays are nonnegative, the noiseless feasible interval is $-z_r \le \theta \le z_f$. Instrument/capture uncertainty widens its endpoints. A narrower guarantee requires measured/ascribed delay and asymmetry bounds, independent calibration, or an independent output-reference comparison. Reciprocity alone does not establish equality of forward and reverse paths, especially with different channels, multipath, RX decoding conditions or motion.

A frequency-correct waveform can still have an arbitrary initial phase and wrong epoch. A divided output can add an ambiguous period or divider phase. The implementation must retain an immutable measurement counter and provenance while separately controlling the output epoch, controlled phase alignment or divider/delay actuator. Resetting the measurement counter to make phase appear correct would invalidate the event history and cannot be an alignment method. A monotonic software API may require slew-only correction; an electrical actuator must characterize any supported step, glitch suppression and settling interval.

The raw-reference domain is not automatically UTC. A future capability record must identify timescale, origin, epoch, leap handling, discontinuity and source quality. Subnanosecond fit scatter cannot repair a wrong second or an unsupported reference domain. Public GM authorization, source quality and this diagnostic pair's numerical convergence remain separate gates.

10 Proposed time-aware PI design

10.1 Detector and actuator units

Consider a new actuator controlling fractional frequency $u = (f - f_0)/f_0$, dimensionless, with phase error $e = T_{\text{ref}} - T_{\text{out}}$ in seconds. If the uncontrolled relative-rate mismatch is δ, the ideal phase dynamics are $\dot{e} = \delta - u$. A proposed feed-forward plus PI law is

$$u^* = \widehat{\delta} + k_p e + k_i I, \qquad \dot{I} = e, \qquad [k_p] = \mathrm{s}^{-1}, \quad [k_i] = \mathrm{s}^{-2}. \tag{7}$$

Here I has units s^2. With a constant mismatch, perfect feed-forward and an unsaturated instantaneous actuator, $\ddot{e} + k_p\dot{e} + k_i e = 0$, giving $\omega_n = \sqrt{k_i}$ and $\zeta = k_p/(2\sqrt{k_i})$. The ideal open loop is $(k_p + k_i/s)/s$. These quantities describe this normalized plant; a real actuator's gain, latency, filtering and inner loop change its poles and margins.

A voltage-controlled device has calibrated $K_f = df/dV$ in Hz/V and $K_y = K_f/f_0$ in V^{-1}. Voltage gains are $K_P = k_p/K_y$ in V/s and $K_I = k_i/K_y$ in $\mathrm{V/s}^2$. A digital-trim device instead supplies K_y in fractional-frequency/code, often represented as ppb/code. One ppb accumulates one ns of phase per second. A carrier detector measured in radians requires the conversion $\phi = 2\pi f_0 e$ and its detector gain; inserting a timestamp-ns gain into a radian PLL is dimensionally wrong.

10.2 Sampling, link delay and stability

Use actual associated hardware observation intervals T_k, not USB arrivals or a fixed assumed packet period. Preserve physical measurement time and availability time separately. A phase/skew state can propagate from observation to actuation through

$$\begin{bmatrix} e \\ \delta \end{bmatrix}_a^- = \begin{bmatrix} 1 & \tau_k \\ 0 & 1 \end{bmatrix} \begin{bmatrix} e \\ \delta \end{bmatrix}_m^+ - \begin{bmatrix} \int_{t_{m,k}}^{t_{a,k}} u_{\text{applied}}(t)\,dt \\ 0 \end{bmatrix}, \qquad \tau_k = t_{a,k} - t_{m,k}, \tag{8}$$

where the integral contains the known commands actually applied between measurement and availability. Uncertainty propagates under an explicitly tested process-noise model. Unknown applied commands contribute uncertainty rather than a fabricated correction. Equation 8 is an original simple constant-skew design model. It does not retroactively change the causal error in Equation 1. Asynchronous estimation can treat variable intervals, but its assumptions and uncertainty calibration need validation [5].

For an explicit-Euler integrator $I_{k+1} = I_k + Te_k$ and a held command $u_k = k_p e_k + k_i I_k$ with zero delay and no saturation, the homogeneous state matrix is

$$A(T) = \begin{bmatrix} 1 - k_p T & -k_i T \\ T & 1 \end{bmatrix}, \quad \lambda^2 - (2 - k_p T)\lambda + 1 - k_p T + k_i T^2 = 0. \tag{9}$$

The second-order Jury inequalities reduce to $k_i > 0$, $k_p > k_i T$ and $4 - 2k_p T + k_i T^2 > 0$. They apply to this stated update ordering only. Backward Euler or use of the just-updated integral gives a different matrix. Variable intervals, command delay, quantization and actuator dynamics require analysis of the augmented state and time-ordered products, rather than claiming that individual stable matrices prove arbitrary-switching stability.

An illustrative, unimplemented choice $\omega_n = 0.02$ rad/s, $\zeta = 1/\sqrt{2}$ gives $k_p = 0.028284$ s^{-1} and $k_i = 0.0004$ s^{-2}. At $T = 0.25$ s, $k_p T = 0.007071$ and $k_i T^2 = 0.000025$, satisfying the ideal conditions. The approximate ideal 2% settling scale $4/(\zeta\omega_n)$ is 283 s. A 0.7425 s delay contributes $\omega_n \tau = 0.01485$ rad at ω_n; this is not a computed crossover phase margin. The example shows why ten seconds of lock dwell is not a loop settling-time proof. No gain set is recommended before identifying the actual plant and observed interval/delay distributions.

Reducing bandwidth suppresses measurement noise but allows more oscillator wander during thermal change and dropout. Increasing bandwidth accelerates correction while admitting path jitter and reducing delay margin. A step/sinusoidal perturbation experiment must measure gain, damping and actual bandwidth with the intended inner synthesizer and scheduling path. Acquisition and maintenance may use different gains with bumpless state transfer and explicit maximum commands. They must not tune on validation tails until those tails disappear.

10.3 Saturation, quantization and anti-windup

Commands obey calibrated voltage/code limits, maximum fractional rate and phase slew, plus actuator rate-of-change and latency constraints. Represent the integral's frequency contribution as $v = k_i I$. A proposed tracking anti-windup law is

$$\dot{v} = k_i e + k_{aw}(u_{\text{applied}} - u^*), \qquad [k_{aw}] = \text{s}^{-1}. \tag{10}$$

Use the actual applied command after all clamps, slew limits and inner-actuator effects; an acknowledgment of a requested code is insufficient if the physical plant is still slewing. Conditional integration is an alternative when integration would drive farther into saturation. These approaches follow the primary anti-windup treatment of Åström and Rundqwist [13]; the normalized equation here is our proposed design. Invalid measurements freeze feedback integration or invoke a specified holdover predictor. A long gap must not create many fictitious updates of a stale error. Origin/epoch change resets or transfers state under a bounded alignment policy.

For a code spacing Δu, one unresolved command step accumulates $\Delta u\, h$ seconds over holdover h. Timestamp quantization, DAC/code noise, nonlinear gain, deadband and phase noise need separate budgets. Dither might improve average rate but adds modulation/phase noise; it requires its own measured output specification. White-noise quantization formulas are illustrative only when independence and distribution are justified.

11 Tunable hardware: concrete extension, absent today

11.1 VCXO, electrically trimmable OCXO and DCO

The proposed path is: associated physical radio timestamps → calibrated phase/skew estimate → time-aware PI and command limiter → calibrated DAC/voltage or digital trim → oscillator and divider/output → independent output observation. The immutable measurement counter provides the event history; an output phase/delay mechanism handles epoch alignment separately. A controller may retain a software map for timestamp services even while disciplining a physical clock, but those two actuators need separate states and evidence.

Microchip describes a VCXO's electrically varied crystal loading; an OCXO stabilizes temperature and needs an exposed electrical-frequency-control option to participate in this loop [14]. An oven does not itself identify phase, remove path bias or guarantee externally tunable frequency. Warm-up, heater energy, aging and control bandwidth enter the qualification. "DCO" must specify whether it controls a crystal, a digitally compensated resonator or a numerical oscillator; the interface name is not the plant model.

Abracon's APVV datasheet specifies a positive-pull analog VCXO with a 0–3.3 V control range and a ± 100 ppm pull option, plus nonzero linearity/aging specifications [15]. As an explicitly idealized local linear example, a 200 ppm excursion over 3.3 V gives $K_y \approx 60{,}606$ ppb/V. A 16-bit 3.3 V DAC's ideal spacing gives approximately 3.052 ppb/code, or 3.052 ns/s accumulated phase per code. These calculated figures are not a measured APVV transfer function or a chosen board design. Actual gain, polarity, range, midpoint, nonlinearity, hysteresis, temperature response, DAC reference noise and settling must be calibrated for the installed unit. DAC nominal bit count alone is not an effective-resolution or stability certificate.

11.2 Clean-up synthesizer and numerical oscillator

A tunable synthesizer provides a different actuator from crystal pulling. Skyworks' Si5341/40 manual exposes fractional-divider frequency adjustment and configuration-dependent phase-step behavior; frequency stepping does not automatically align the output epoch [16]. The outer wireless loop must account for bus delay, update timing, inner PLL bandwidth, divider state and any permitted phase adjustment. Its gain is frequency-plan/code specific.

The AD9545 datasheet describes DPLL/NCO operation, programmable bandwidth, frequency/phase lock detection, phase slew and holdover frequency history [17]. A digital NCO still depends on its real system-clock reference; its computed word does not eliminate that reference's drift during holdover. If a clean-up device has an active inner loop, the measured composite plant replaces the ideal instantaneous u in Equation 7. Two uncoordinated integral loops can change dynamics substantially. Outer-loop tuning must verify total delay and avoid claiming the inner device's lock flag as wireless output qualification.

External two-tone clock recovery, numerical timestamp discipline, tunable quartz control and RF carrier coherence are therefore four distinct capabilities. A node may support one or several. A radio derived from a disciplined reference can still add synthesizer/divider phase ambiguity and antenna/channel phase, so even clock-frequency lock does not prove coherent RF transmission. The current experiment supports only the numerical-map member of this set.

11.3 Dropout and bounded holdover

A model-based holdover policy propagates output uncertainty rather than continuing to advertise its last lock. With a qualified initial phase uncertainty U_0, rate uncertainty U_y and rate-change bound U_g, a conservative planning relation is

$$U(h) \le U_0 + U_y h + \tfrac{1}{2} U_g h^2 + U_{\text{act}}(h) + U_{\text{path}}(h). \tag{11}$$

In ns/ppb units, U_y ppb times h seconds gives ns; U_g ppb/s times $h^2/2$ gives ns. These terms are bounds only after qualification. Temperature/aging response, command uncertainty, epoch discontinuity and path uncertainty cannot be replaced by a recent residual RMS. The current console's 50 ppm aging scenario is explicitly unqualified.

During dropout, freeze or predict the trim under a documented uncertainty model; withdraw locked status and fall back when the bound exceeds the operating requirement. Recovery measures the whole missing-reference interval, command saturation, possible phase excursion, reacquisition and subsequent dwell. A current three-second freshness guard is an algorithm parameter, not a measured blackout guarantee. NIST's frequency-stability guidance distinguishes phase/frequency statistics and requires deliberate treatment of gaps and irregular samples [18]. Allan/TDEV studies should use independently measured output on a documented regular grid, rather than silently regularizing these irregular packet errors.

12 Optional supported-device operating tier

The evidence supports retaining `SOFTWARE_PLL` as an optional highest algorithmic operating tier for a supported device/profile, subject to the native causal-history, actual-update, permission and epoch gates. Its `LOCKED` state expresses virtual phase/frequency convergence with the intended purpose of driving fine crystal correction on capable hardware. An implementation should expose algorithm state, detector/control profile, actuator support and engagement, fresh observed state, reference domain and output qualification together. The current experiment validates the software-map case; calibrated physical-actuator readiness and an electrical precision grade require their own tests. A simpler passive client continues to consume the existing time service without implementing the tier.

Table 5: Proposed capability/quality contract. Entries describe negotiated support and qualification evidence, not newly implemented wire fields.

Capability	Required meaning and downgrade condition
Reference provenance	Origin, domain/timescale, epoch/session, authority and lease, discontinuity rules; change invalidates old convergence.
Detector	Event/edge, PHY/channel, capture grid and counter mapping, exact original/follow-up association, calibration identity and its applicable conditions.
Actuator	Software map, crystal voltage/trim, synthesizer/NCO or external recovered reference; gain units, range, resolution, slew, latency, phase/epoch mechanism and acknowledged applied state.
Native convergence	Algorithmic virtual lock for fine phase/frequency correction; actual update age/count, both causal histories, phase remaining, signed trend, dwell/hysteresis and explicit fault reasons; actuator engagement reported separately.
Output and uncertainty	Numerical query, timer compare, electrical PPS/clock or RF carrier; qualified reference/path/output uncertainty, calibration validity and permitted holdover horizon, otherwise UNKNOWN.
Fallback	Preserve public source authorization and monotonic-time contract; withdraw premium state on loss of gate/lease/origin and report a supported coarse/holdover/unavailable state.

Negotiation must be versioned, bounded and backward compatible. An unsupported peer must ignore

optional refinement safely rather than infer qualified precision from a rate number. Resource-heavy detector/calibration/output adapters belong to capable nodes. Multi-follower feedback requires an explicit contention budget and advancing-sequence semantics; routine extra RF messages cannot be assumed free. Direct and forwarded references need separately supported physical-stamp and root/residence provenance. No current measurement qualifies a relay or an arbitrary new hardware adapter.

Promoting an electrical-output tier requires a separate qualification record: unit, installed image, actuator calibration, detector/path profile, source/epoch, actual output interface, instrument uncertainty, full coverage and maximum total error. Private pair feedback cannot change public GM authority, manufacture source quality or bypass the authenticated control lease. A malicious or faulty peer's plausible RMS is not independent evidence. Formal Bluetooth qualification and IEEE conformance remain separate outcomes.

For future BLE/PTS dual operation, PTS remains off until each actual W12 unit's held delivered time is qualified within $\pm 500\,\mathrm{ns}$ through the whole actual blackout plus age, switching and synchronization margins. Actual BLE listening must be at most 50%, preferably coordinated. Neither a 1 μs hold estimate nor P95/P99 nor this raw T096 pair's recent RMS satisfies that gate. TC03's no-battery capability and factory-backup exception do not supply a measured holdover source.

12.1 Proposed SoftPLL update after IEEE 1588 and LoRa restoration

The operator requests a further reviewable update after the IEEE 1588 GM work and LoRa restoration are integrated and tested. This manuscript proposes that sequence; it records no additional firmware deployment. The follower header would change from the installed `PLL LOCK` to `SoftPLL`. Before native lock, the existing `PAIR RMS` presentation remains. During native lock, a blue inline lock accompanies the remaining `CORR` value. After zero correction holds for at least five hardware seconds and four advancing originals while all native gates remain satisfied, the value becomes green `LOCKED`. Nonzero correction clears the green confirmation; lost algorithmic lock restores the earlier RMS presentation. A public GM continues to show PPB. These rules apply equally to TC01, TC02 and TC03 when acting as followers. The displayed lock is virtual algorithmic convergence, with crystal actuation conditional on advertised and validated support.

Before formalizing the extension, the proposed work makes provenance, reference changes and reboot policy consistent with the completed GM/source-selection model; keeps BLE and LoRa estimates separate; negotiates detector/actuator profiles and honest UNKNOWN quality; measures assembly latency, continuity and observer effects; and compares acquisition/maintenance cadence with multiple followers. Lock thresholds and servo gains belong to validated profiles rather than one universal crystal-control constant. A physical trim adapter additionally needs calibrated gain and polarity, applied-command/delay accounting, anti-windup, phase/epoch alignment and uncertainty-bounded holdover. Existing public wire formats and controller guards remain intact. The repository's companion post-integration proposal specifies the UI states, implementation sequence and acceptance tests.

13 Discriminating validation plan

The next experiment should identify mechanisms and complete the missing acceptance evidence, rather than merely seek another favorable lock snapshot. Freeze installed identities, capability/calibration profiles, control budgets, logger configuration and a prospective scoring plan before collecting new measurements.

1. Calibrate the detector and actuator independently. Associate physical stamps; measure transmission, reception, cable and antenna delay and directional asymmetry under the intended PHY, channels, signal levels and geometry. Measure oscillator gain and polarity, command range, quantization, latency, hysteresis, temperature and aging response against an independent frequency reference. Keep training and validation separate.

2. Connect the actual electrical output to an independently calibrated time-interval instrument or suitable logic/analyzer system. Record reference edges, pad edges, cable delays, resolution and expanded uncertainty, with a regular prospectively defined opportunity grid. Retain unavailable outputs, startup, end censoring, dropped records and all maxima. Shared-counter predictions alone cannot complete this step.

3. Test power-cold and warm restart, source/epoch change, acquisition, steady maintenance, deliberate packet loss and completion delay, saturation, full dropout/holdover/recovery, temperature and coexistence. Perturb phase and frequency separately to identify bandwidth/damping. Validate software-map and physical-actuator behavior separately, including monotonic output and divider alignment.

4. Measure completion and availability at each stage: emitted original, physical reception, follow-up association, permitted control update and delivered output. Measure actual RF/listening residence, CPU busy time, queue drops and whole-device energy. Run matched logging-enabled/disabled treatments with external observation to quantify the observer effect.

5. Repeat fixed and adaptive cadences with multiple followers and supported hardware profiles. Predefine lock/holdover/precision withdrawal, qualify total-error maxima over the entire intended service interval and use repeated/block-aware analysis for correlated data. Seal every failure and incomplete run after all writers stop.

This plan is an extension proposal. No new device access, tuning, output experiment or energy measurement was performed for the manuscript.

14 Interpretation and conclusion

The native failure is a useful result: plausible timestamp noise did not establish readiness, and host probes exposed a coherent-age permission mechanism capable of repeated invalidation. Individual native V3 withdrawals remain unattributed. Later V4 captures with the coherence repair and explicit causal actuator establish native convergence while preserving radio/fault guards. UI5 separates local native lock, remaining numerical phase and settled presentation; peer reports and rounded RMS retain weaker meanings.

The quiet and delayed-join capture distributions establish useful conditional software-map alignment to the raw reference under the recorded profiles. The final first capture's 231.105 ns RMS and 1413 ns maximum, the simultaneous three-unit failure, retained long gaps and logging drops constrain the result. The second final capture's 79.896 ns RMS and last 84 ns recent display do not erase them. The data support neither a pooled highest-precision bound nor uninterrupted lock duration, physical crystal/carrier discipline, calibrated UTC, field distance/restart qualification or resource savings.

A capability-scoped `SOFTWARE_PLL` tier can communicate virtual algorithmic lock and the intended readiness for fine crystal discipline on supported hardware, while keeping actuator engagement, UNKNOWN physical quality and safe fallback explicit. The current boards demonstrate the software-map control case. A future tunable device needs a calibrated detector-to-actuator chain, dimensional gains, time/delay-aware control, anti-windup, independent phase/epoch alignment, uncertainty growth and independent electrical validation. The mathematics supplies a design framework; the manufacturer's tuning features supply possible actuators; only prospective physical experiments can supply their performance claim.

Reproducibility, data and assistance disclosure

The local evidence manifest links the nine separately sealed cohorts, final/native indexes, capture/epoch metrics, immutable method hashes and indexed raw ranges. The figure directory contains the derivation script, full-record

scientific exports, per-capture metrics and event-referenced data. The manuscript embeds its vector plots and bibliography directly, so the standalone editor needs no external project dependencies. Private raw console logs are not uploaded or released by this draft. Same-host seal integrity is distinct from independent backup.

Jonathan Yantis (Engineer) supplies the hardware, original project and experimental direction and is the human author. OpenAI Codex assisted with primary-source review, offline seal/metric auditing, event-referenced extraction, the figure derivation, controller/unit analysis, manuscript drafting and LaTeX preparation. The operator and reviewers must assess the scientific claims before submission; AI assistance is disclosed as assistance rather than human authorship, following the project's contribution record. This is a local review draft, not a submitted or externally published paper.

References

[1] J. Yantis, "Passive Precision Time over BLE: Prospective Edge Calibration, Causal Publication, and Service Availability," local preprint, 2 Oct. 2026. Frozen companion manuscript: `paper/ptob-preprint.tex`; hash in Ref. [19].

[2] D. Mills, J. Martin, J. Burbank and W. Kasch, *Network Time Protocol Version 4: Protocol and Algorithms Specification*, RFC 5905, Jun. 2010, Secs. 11.3 and 12. https://www.rfc-editor.org/rfc/rfc5905.html.

[3] R. Cochran and LinuxPTP contributors, primary PI servo implementation, `pi.c`, and servo configuration documentation, accessed 3 Oct. 2026. https://github.com/richardcochran/linuxptp/blob/master/pi.c; https://www.linuxptp.org/documentation/default/.

[4] P. Tosato, D. Macii, D. Fontanelli, D. Brunelli and D. Laverty, "A Software-based Low-Jitter Servo Clock for Inexpensive Phasor Measurement Units," 2019, Sec. III. https://arxiv.org/abs/1902.09912.

[5] N. M. Freris, V. S. Borkar and P. R. Kumar, "Model-based clock synchronization protocol," arXiv:1311.6914, inspected version dated 8 Sep. 2021, Secs. IV and VI. https://arxiv.org/abs/1311.6914.

[6] J. C. Park and G. Wolverton, "Wireless Time and Frequency Lock Loop," US Patent 10,856,242 B2, 1 Dec. 2020, Figs. 3–7 and accompanying description. https://patents.google.com/patent/US10856242B2/en.

[7] O. Abari, H. Rahul and D. Katabi, "Poster: Clock Synchronization for Distributed Wireless Protocols at the Physical Layer," ACM MobiCom, 2014. https://people.csail.mit.edu/rahul/papers/clocksync-mobicom2014.pdf.

[8] K. Alemdar, D. Varshey, S. Mohanti, U. Muncuk and K. Chowdhury, "RFClock: Timing, Phase and Frequency Synchronization for Distributed Wireless Networks," ACM MobiCom, 2021, doi:10.1145/3447993.3448623, Secs. 4–6. https://genesys-lab.org/papers/RFCLOCK_MOBICOM2021.pdf.

[9] S. Sridhar, P. Misra and J. Warrior, "CheepSync: A Time Synchronization Service for Resource Constrained Bluetooth LE Advertisers," arXiv:1501.06479, 2015, revised 2016. https://arxiv.org/abs/1501.06479.

[10] F. Asgarian, *Low-Power Low-Error Wireless Sensor Network for Health and Usage Monitoring Systems*, Ph.D. dissertation, University of Michigan, 2022, Ch. 2, printed pp. 22–24 and 36–38. Institutional primary source: https://backend.production.deepblue-documents.lib.umich.edu/server/api/core/bitstreams/1818d96c-dce0-4612-97a7-2bf5d074174c/content.

[11] Nordic Semiconductor, "Wireless timer synchronization among nRF5 devices," primary reference design, accessed 3 Oct. 2026. https://devzone.nordicsemi.com/guides/short-range-guides/b/bluetooth-low-energy/posts/wireless-timer-synchronization-among-nrf5-devices.

[12] R. Ohara et al., "MicroSync: Sub-Micro Second Accuracy Wireless Time Synchronization Service," *IEEE Access*, vol. 12, pp. 124478–124494, 2024, doi:10.1109/ACCESS.2024.3446668, Sec. V. https://da.lib.kobe-u.ac.jp/da/kernel/0100491658/0100491658.pdf.

[13] K. J. Åström and L. Rundqwist, "Integrator Windup and How to Avoid It," *American Control Conference*, 1989, pp. 1693–1698, doi:10.23919/ACC.1989.4790464. Author-paper copy: https://cse.lab.imtlucca.it/~bemporad/teaching/controllodigitale/pdf/Astrom-ACC89.pdf.

[14] Microchip Technology, "Voltage-Controlled Crystal Oscillators (VCXO)," official component documentation, accessed 3 Oct. 2026. https://www.microchip.com/en-us/products/clock-and-timing/components/oscillators/vcxo.

[15] Abracon, *APVV CMOS VCXO Datasheet*, revision 01.17.13, p. 1. https://abracon.com/Oscillators/APVV.pdf.

[16] Skyworks Solutions, *Si5341/Si5340 Rev D Family Reference Manual*, rev. 1.3, 7 Dec. 2021, Sec. 6. https://www.skyworksinc.com/-/media/Skyworks/SL/documents/public/reference-manuals/Si5341-40-D-RM.pdf.

[17] Analog Devices, *AD9545 Quad Input, 10-Output, Dual DPLL/IEEE 1588 Synchronizer and Jitter Cleaner Datasheet*, rev. C, DPLL/NCO, holdover, lock and phase-slew sections, pp. 94, 99, 104–105. https://www.analog.com/media/en/technical-documentation/data-sheets/AD9545.pdf.

[18] W. J. Riley, *Handbook of Frequency Stability Analysis*, NIST Special Publication 1065, 2008, Secs. 10.4–10.10. https://nvlpubs.nist.gov/nistpubs/Legacy/SP/nistspecialpublication1065.pdf.

[19] J. Yantis with OpenAI Codex assistance, local research evidence and primary-source review, 3 Oct. 2026: `paper/oscillator-pll-evidence-20261003.json`, `paper/provenance/oscillator-pll-literature-20261003.json`, and `paper/figures/oscillator-pll-20261003/derive.py`. Private archive paths, seals, metric definitions, exact denominators and byte-range locators are recorded there.